AxleMetrics

REPORT DATE
2026-01-11

# HIGHWAY PAVEMENT DESIGN REPORT

Flexible Pavement Simplified AASHO ESAL Factor

## COMPANY INFORMATION

COMPANY
Kwame Nkrumah University of Science and Technology

PHONE
0201603517

ADDRESS
Adom Drive

## PROJECT SUMMARY

PROJECT NAME
Kumasi-Accra

PREPARED BY
Micah Richards

## CALCULATION PARAMETERS

COMPUTATION METHOD
Simplified AASHO Equations

STANDARD LOAD
8164 kg

LOAD EQUIVALENCY FACTOR (C)
4.5

## CALCULATED RESULTS

Summary of axle types and their respective Equivalent Single Axle Load (ESAL) factors.

| AXLE TYPE | AVERAGE ESAL FACTOR |
|---|---|
| 7-axle trailers | 68.1 496 |
| 6-axle trailers | 3889.2761 |
| 3-axle trailers | 77.8329 |
| 4-axle trailers | 797.4604 |

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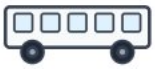
## VEHICLE CLASSIFICATION REFERENCE

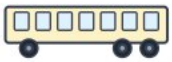
**Passenger Car**
PC • 2 axles
2 axles, 4 tires

**Small Buses**
SB • 2 axles
2 axles



**Medium Buses**
MB • 2 axles
2 axles

**Large Buses (2-axle)**
LB2 • 2 axles
2 axles



**Large Buses (3-axle)**
LB3 • 3 axles
3 axles

**Light Trucks**
LT • 2 axles
2 axles, pickup/van

**Medium Trucks**
MT1 • 2 axles
2 axles

**Heavy Trucks**
HT5 • 3 axles
3+ axles

**3-axle trailers**
3T1 • 3 axles
Tractor + trailer

**4-axle trailers**
4T1 • 4 axles
Tractor + trailer

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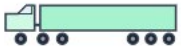
**5-axle trailers**
5T1 • 5 axles
Tractor + trailer



**6-axle trailers**
6T1 • 6 axles
Tractor + trailer

**7-axle trailers**
7T1 • 7 axles
Tractor + trailer

**8-axle trailers**
8T1 • 8 axles
Tractor + trailer

**9-axle trailers**
9T1 • 9 axles
Tractor + trailer

**10-axle trailers**
10T • 10 axles
Tractor + trailer

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